

Panduan Setup Web dan Hadoop

Smart Clothesline IoT System - Big Data Analysis Feature

Disusun untuk implementasi branch: feature/analysis-hadoop-big-data

Tanggal: 2026-06-02

Ringkasan:

- Setup dibagi menjadi dua jalur: Web/Next.js dan Hadoop Cluster.
- Fondasi GitHub develop saat ini masih memiliki Big Data scaffold, bukan implementasi Hadoop penuh.
- Target implementasi adalah pipeline Firestore sensor_data -> CSV -> HDFS -> MapReduce -> JSON -> fitur Analysis.
- Hadoop berjalan sebagai batch analytics. MQTT tetap dipakai untuk realtime telemetry/control.

Area	Status develop terbaru	Implikasi
package.json	Belum ada script bigdata:* atau test:bigdata.	Perlu ditambah di branch Big Data.
/analytics	Sudah ada operational analytics.	Tambahkan section/tab Hadoop Batch Analytics tanpa rewrite total.
/big-data	Masih sample page sederhana.	Upgrade menjadi halaman detail Hadoop report.
BIG_DATA_ROADMAP.md	Menjadi roadmap canonical.	Jadikan acuan utama implementasi.
export script	Masih scaffold/sample mode.	Perbaiki kontrak CSV dan fallback sample.

Daftar Isi

Gunakan heading di dokumen ini sebagai navigasi. Versi PDF dibuat untuk dibaca langsung dan dicari dengan fitur search.

1. Setup Web / Project Next.js

1.1 Clone repo atau masuk repo existing

```
cd /d/PBL
```

```
git clone https://github.com/afadlih/smart-clothesline-iot-system.git
```

```
cd smart-clothesline-iot-system
```

Jika repo sudah ada:

```
cd /d/PBL/smart-clothesline-iot-system
```

1.2 Ambil branch develop

```
git fetch origin --prune
```

```
git checkout develop
```

```
git pull origin develop
```

```
git branch --show-current
```

Expected output:

```
develop
```

1.3 Install dependency

```
npm ci
```

Jika npm ci error karena lockfile mismatch, gunakan sementara:

```
npm install
```

1.4 Validasi baseline

```
npm run lint
```

```
npm run typecheck
```

```
npm run test:telegram
```

```
npm run test:architecture
```

```
npm run build
```

```
npm run validate
```

Baseline validate di develop saat ini:

```
validate = lint + typecheck + test:telegram + test:architecture + build
```

1.5 Setup .env.local

Buat .env.local. Jangan commit file ini.

```
NEXT_PUBLIC_FIREBASE_API_KEY=...
```

```
NEXT_PUBLIC_FIREBASE_AUTH_DOMAIN=...
```

```
NEXT_PUBLIC_FIREBASE_PROJECT_ID=...
```

NEXT_PUBLIC_FIREBASE_STORAGE_BUCKET=...
NEXT_PUBLIC_FIREBASE_MESSAGING_SENDER_ID=...
NEXT_PUBLIC_FIREBASE_APP_ID=...

NEXT_PUBLIC_MQTT_BROKER_URL=wss://<hivemq-cloud-host>:8884/mqtt
NEXT_PUBLIC_MQTT_USERNAME=<mqtt-user>
NEXT_PUBLIC_MQTT_PASSWORD=<mqtt-password>

TELEGRAM_BOT_TOKEN=...
TELEGRAM_CHAT_ID=...
APP_BASE_URL=http://localhost:3000

Catatan: NEXT_PUBLIC_ terlihat di browser. Gunakan credential dengan permission terbatas.*

1.6 Jalankan web app

npm run dev

Buka:

http://localhost:3000
http://localhost:3000/analytics
http://localhost:3000/big-data

1.7 Buat branch Big Data

git checkout develop
git pull origin develop
git checkout -b feature/analysis-hadoop-big-data
git push -u origin feature/analysis-hadoop-big-data

Scope branch ini khusus untuk Hadoop Big Data analytics MVP di fitur Analysis. Jangan campur dengan MQTT realtime, Telegram behavior, schedule runtime, auth, atau dashboard command.

2. Struktur Big Data yang Perlu Dibuat di Web Project

hadoop/
 README.md
 pom.xml
 src/main/java/com/smartclothesline/bigdata/DailySensorSummaryJob.java
 src/main/java/com/smartclothesline/bigdata/RainEventAggregationJob.java
 src/main/java/com/smartclothesline/bigdata/common/SensorCsvRecord.java
 src/main/java/com/smartclothesline/bigdata/common/SafeNumber.java
 sample/
 sensor_input.csv
 daily-summary.expected.txt
 rain-events.expected.txt

scripts/bigdata/

generate-sample-sensor-csv.mjs
build-hadoop-jobs.mjs
run-local-mapreduce-sample.mjs
deploy-to-hadoop-cluster.mjs
fetch-hadoop-output.mjs
export-firestore-sensor-data.mjs
convert-mapreduce-output-to-json.mjs
bigdata-contract.test.mjs

src/services/
BigDataReportService.ts

public/bigdata-reports/
daily-summary.sample.json
rain-events.sample.json

docs/
HADOOP_ZEROTIER_CLUSTER_RUNBOOK.md
HADOOP_DATA_DICTIONARY.md
HADOOP_DEMO_SCRIPT.md

2.1 CSV contract final

deviceId,receivedAt,date,hour,temperature,humidity,light,rain,status,mode,source

Contoh:

wokwi-default,1779167140069,2026-05-19,14,30.7,73,1200,true,CLOSED,AUTO,WOKWI
wokwi-default,1779167200069,2026-05-19,14,31.1,74,1400,true,CLOSED,AUTO,WOKWI
esp32-demo-01,1779167260069,2026-05-19,15,29.8,70,6200,false,OPEN,AUTO,DEVICE

Field	Tipe	Keterangan
deviceId	string	ID device, misalnya wokwi-default atau esp32-demo-01.
receivedAt	number	Epoch milliseconds untuk waktu server/browser menerima data.
date	YYYY-MM-DD	Tanggal agregasi.
hour	0..23	Jam agregasi.
temperature	number	Suhu.
humidity	number	Kelembapan.
light	number	Nilai cahaya.
rain	boolean	true jika hujan terdeteksi.
status	enum	OPEN/CLOSED/MOVING/ RESTARTING/FAULT/UNKNOWN.
mode	enum	AUTO/MANUAL/SCHEDULE/ UNKNOWN.

source	string	WOKWI/DEVICE/STATUS_TOPIC/ SENSOR_FALLBACK/UNKNOWN.
--------	--------	--

2.2 MapReduce Jobs

DailySensorSummaryJob

Group by:

date,deviceId

Output:

date,deviceId<TAB>totalRecords,avgTemperature,avgHumidity,avgLight,rainCount,openCount,closedCount

Example:

2026-05-19,wokwi-default 1440,30.40,72.80,6200.00,18,820,620

RainEventAggregationJob

Group by:

date,deviceId,hour

Output:

date,deviceId,hour<TAB>totalRecords,rainCount,rainPercentage,avgLight,avgHumidity

Example:

2026-05-19,wokwi-default,14 60,12,20.00,2400.50,81.20

2.3 Package scripts yang ditambahkan

```
{
  "bigdata:sample": "node scripts/bigdata/generate-sample-sensor-csv.mjs",
  "bigdata:build": "node scripts/bigdata/build-hadoop-jobs.mjs",
  "bigdata:local": "node scripts/bigdata/run-local-mapreduce-sample.mjs",
  "bigdata:deploy": "node scripts/bigdata/deploy-to-hadoop-cluster.mjs",
  "bigdata:fetch": "node scripts/bigdata/fetch-hadoop-output.mjs",
  "bigdata:convert": "node scripts/bigdata/convert-mapreduce-output-to-json.mjs",
  "test:bigdata": "node --test scripts/bigdata/bigdata-contract.test.mjs"
}
```

Update validate:

```
{
  "validate": "npm run lint && npm run typecheck && npm run test:telegram && npm run test:architecture && npm run test:bigdata && npm run build"
}
```

Jangan masukkan `bigdata:build` ke `validate` karena `Maven/Hadoop` belum tentu ada di `CI`.

3. Setup Hadoop Cluster dari Awal

Kondisi tim: 1 NameNode utama, 3 DataNode, dan ZeroTier sebagai jaringan privat antar VM/laptop.

Node	Role	Service
hadoop-namenode	Master	NameNode, ResourceManager, JobHistoryServer, run command HDFS/YARN/MapReduce.
hadoop-datanode1	Worker	DataNode, NodeManager.
hadoop-datanode2	Worker	DataNode, NodeManager.
hadoop-datanode3	Worker	DataNode, NodeManager.

3.1 Hostname dan /etc/hosts

Contoh IP ZeroTier. Ganti sesuai IP tim.

```
10.147.17.10 hadoop-namenode
```

```
10.147.17.11 hadoop-datanode1
```

```
10.147.17.12 hadoop-datanode2
```

```
10.147.17.13 hadoop-datanode3
```

Set hostname:

```
# NameNode
```

```
sudo hostnamectl set-hostname hadoop-namenode
```

```
# DataNode 1
```

```
sudo hostnamectl set-hostname hadoop-datanode1
```

```
# DataNode 2
```

```
sudo hostnamectl set-hostname hadoop-datanode2
```

```
# DataNode 3
```

```
sudo hostnamectl set-hostname hadoop-datanode3
```

Di semua VM, edit /etc/hosts:

```
sudo nano /etc/hosts
```

```
10.147.17.10 hadoop-namenode
```

```
10.147.17.11 hadoop-datanode1
```

```
10.147.17.12 hadoop-datanode2
```

```
10.147.17.13 hadoop-datanode3
```

Test koneksi:

```
ping -c 3 hadoop-namenode
```

```
ping -c 3 hadoop-datanode1
```

```
ping -c 3 hadoop-datanode2
```

```
ping -c 3 hadoop-datanode3
```

3.2 User Hadoop

```
sudo adduser hadoop
sudo usermod -aG sudo hadoop
su - hadoop
```

3.3 SSH tanpa password dari NameNode ke DataNode

```
# Jalankan di NameNode sebagai user hadoop
ssh-keygen -t rsa -P "" -f ~/.ssh/id_rsa
cat ~/.ssh/id_rsa.pub >> ~/.ssh/authorized_keys
chmod 700 ~/.ssh
chmod 600 ~/.ssh/authorized_keys
```

```
ssh-copy-id hadoop@hadoop-datanode1
ssh-copy-id hadoop@hadoop-datanode2
ssh-copy-id hadoop@hadoop-datanode3
```

```
ssh hadoop@hadoop-datanode1 hostname
ssh hadoop@hadoop-datanode2 hostname
ssh hadoop@hadoop-datanode3 hostname
```

3.4 Install Java dan tools di semua VM

```
sudo apt update
sudo apt install -y openjdk-11-jdk wget tar rsync curl nano
```

```
java -version
readlink -f $(which java)
```

JAVA_HOME umum di Ubuntu:

```
/usr/lib/jvm/java-11-openjdk-amd64
```

3.5 Install Hadoop di semua VM

```
cd /tmp
wget https://downloads.apache.org/hadoop/common/hadoop-3.4.2/hadoop-3.4.2.tar.gz
tar -xzf hadoop-3.4.2.tar.gz
sudo mv hadoop-3.4.2 /opt/hadoop
sudo chown -R hadoop:hadoop /opt/hadoop
```

Jika link versi berubah, pakai versi Hadoop stable terbaru yang sama di semua VM.

Tambahkan env di ~/.bashrc semua VM:

```
export JAVA_HOME=/usr/lib/jvm/java-11-openjdk-amd64
export HADOOP_HOME=/opt/hadoop
export HADOOP_CONF_DIR=$HADOOP_HOME/etc/hadoop
export PATH=$PATH:$HADOOP_HOME/bin:$HADOOP_HOME/sbin
source ~/.bashrc
```

hadoop version

3.6 Folder HDFS

Di NameNode:

```
sudo mkdir -p /data/hadoop/hdfs/namenode
sudo chown -R hadoop:hadoop /data/hadoop
```

Di setiap DataNode:

```
sudo mkdir -p /data/hadoop/hdfs/datanode
sudo chown -R hadoop:hadoop /data/hadoop
```

3.7 Konfigurasi Hadoop di NameNode

Semua file ada di /opt/hadoop/etc/hadoop. Setelah selesai, copy ke DataNode.

hadoop-env.sh

```
export JAVA_HOME=/usr/lib/jvm/java-11-openjdk-amd64
```

core-site.xml

```
<configuration>
  <property>
    <name>fs.defaultFS</name>
    <value>hdfs://hadoop-namenode:9000</value>
  </property>
</configuration>
```

hdfs-site.xml

```
<configuration>
  <property>
    <name>dfs.namenode.name.dir</name>
    <value>file:///data/hadoop/hdfs/namenode</value>
  </property>

  <property>
    <name>dfs.datanode.data.dir</name>
    <value>file:///data/hadoop/hdfs/datanode</value>
  </property>

  <property>
    <name>dfs.replication</name>
    <value>3</value>
  </property>

  <property>
    <name>dfs.namenode.http-address</name>
    <value>hadoop-namenode:9870</value>
  </property>
```



```
<property>
  <name>dfs.permissions.enabled</name>
  <value>>false</value>
</property>
</configuration>
```

yarn-site.xml

```
<configuration>
  <property>
    <name>yarn.resourcemanager.hostname</name>
    <value>hadoop-namenode</value>
  </property>

  <property>
    <name>yarn.resourcemanager.webapp.address</name>
    <value>hadoop-namenode:8088</value>
  </property>

  <property>
    <name>yarn.nodemanager.aux-services</name>
    <value>mapreduce_shuffle</value>
  </property>

  <property>
    <name>yarn.nodemanager.env-whitelist</name>

    <value>JAVA_HOME,HADOOP_COMMON_HOME,HADOOP_HDFS_HOME,HADOOP_CONF_DIR,CLASSPATH_PREPEND_DISTCACHE,H
ADOOP_YARN_HOME,HADOOP_HOME,PATH,LANG,TZ</value>
  </property>
</configuration>
```

mapred-site.xml

```
<configuration>
  <property>
    <name>mapreduce.framework.name</name>
    <value>yarn</value>
  </property>

  <property>
    <name>mapreduce.jobhistory.address</name>
    <value>hadoop-namenode:10020</value>
  </property>
```

```
<property>
  <name>mapreduce.jobhistory.webapp.address</name>
  <value>hadoop-namenode:19888</value>
</property>
</configuration>
```

workers

```
hadoop-datanode1
hadoop-datanode2
hadoop-datanode3
```

3.8 Copy config ke DataNode

```
rsync -av /opt/hadoop/etc/hadoop/ hadoop@hadoop-datanode1:/opt/hadoop/etc/hadoop/
rsync -av /opt/hadoop/etc/hadoop/ hadoop@hadoop-datanode2:/opt/hadoop/etc/hadoop/
rsync -av /opt/hadoop/etc/hadoop/ hadoop@hadoop-datanode3:/opt/hadoop/etc/hadoop/
```

3.9 Format NameNode

```
hdfs namenode -format
```

Hanya lakukan sekali saat setup awal. Jangan format ulang setelah data masuk.

3.10 Start HDFS dan YARN

```
start-dfs.sh
start-yarn.sh
mapred --daemon start historyserver
```

Cek proses di NameNode:

```
jps
```

Expected:

```
NameNode
ResourceManager
JobHistoryServer
Jps
```

Cek proses di DataNode:

```
ssh hadoop@hadoop-datanode1 jps
ssh hadoop@hadoop-datanode2 jps
ssh hadoop@hadoop-datanode3 jps
```

Expected:

```
DataNode
NodeManager
Jps
```

3.11 Validasi cluster

```
hdfs dfsadmin -report
```

```
yarn node -list
```

Expected: Live DataNodes = 3 dan NodeManagers = 3.

Web UI:

NameNode UI: `http://<ZEROTIER_IP_NAMENODE>:9870`

YARN UI: `http://<ZEROTIER_IP_NAMENODE>:8088`

History UI: `http://<ZEROTIER_IP_NAMENODE>:19888`

4. Integrasi Web ke Hadoop

4.1 Generate sample CSV

```
npm run bigdata:sample
```

Expected:

```
exports/sample-sensor-data.csv
```

4.2 Build Hadoop JAR

```
npm run bigdata:build
```

Expected:

```
hadoop/target/smart-clothesline-bigdata.jar
```

Jika Maven hanya ada di NameNode:

```
scp -r hadoop hadoop@$HADOOP_NAMENODE_HOST:/tmp/smart-clothesline-hadoop
```

```
ssh hadoop@$HADOOP_NAMENODE_HOST "cd /tmp/smart-clothesline-hadoop && mvn clean package"
```

4.3 Deploy CSV dan JAR ke NameNode

```
npm run bigdata:deploy -- --csv exports/sample-sensor-data.csv --jar hadoop/target/smart-clothesline-bigdata.jar
```

Manual fallback:

```
ssh hadoop@$HADOOP_NAMENODE_HOST "mkdir -p /tmp/smart-clothesline"
```

```
scp exports/sample-sensor-data.csv hadoop@$HADOOP_NAMENODE_HOST:/tmp/smart-clothesline/sensor.csv
```

```
scp hadoop/target/smart-clothesline-bigdata.jar hadoop@$HADOOP_NAMENODE_HOST:/tmp/smart-clothesline/app.jar
```

4.4 Upload CSV ke HDFS

```
hdfs dfs -mkdir -p /input/smart-clothesline
```

```
hdfs dfs -put -f /tmp/smart-clothesline/sensor.csv /input/smart-clothesline/sensor.csv
```

```
hdfs dfs -head /input/smart-clothesline/sensor.csv
```

4.5 Run MapReduce jobs

```
hdfs dfs -rm -r -f /output/smart-clothesline/daily-summary
```

```
hdfs dfs -rm -r -f /output/smart-clothesline/rain-events
```

```
hadoop jar /tmp/smart-clothesline/app.jar com.smartclothesline.bigdata.DailySensorSummaryJob  
/input/smart-clothesline/sensor.csv /output/smart-clothesline/daily-summary
```

```
hadoop jar /tmp/smart-clothesline/app.jar com.smartclothesline.bigdata.RainEventAggregationJob  
/input/smart-clothesline/sensor.csv /output/smart-clothesline/rain-events
```

4.6 Cek hasil MapReduce

```
hdfs dfs -cat /output/smart-clothesline/daily-summary/part-r-00000
```

```
hdfs dfs -cat /output/smart-clothesline/rain-events/part-r-00000
```

Jika output file bernama part-00000, sesuaikan path di command fetch/converter.

4.7 Fetch output ke Web Project

```
npm run bigdata:fetch
```

Manual fallback:

```
ssh hadoop@$HADOOP_NAMENODE_HOST "hdfs dfs -get -f /output/smart-clothesline/daily-summary/part-r-00000 /tmp/smart-clothesline/daily-summary.part"
```

```
ssh hadoop@$HADOOP_NAMENODE_HOST "hdfs dfs -get -f /output/smart-clothesline/rain-events/part-r-00000 /tmp/smart-clothesline/rain-events.part"
```

```
mkdir -p hadoop/output
```

```
scp hadoop@$HADOOP_NAMENODE_HOST:/tmp/smart-clothesline/daily-summary.part hadoop/output/daily-summary.part
```

```
scp hadoop@$HADOOP_NAMENODE_HOST:/tmp/smart-clothesline/rain-events.part hadoop/output/rain-events.part
```

4.8 Convert output ke JSON

```
node scripts/bigdata/convert-mapreduce-output-to-json.mjs --type daily-summary --input hadoop/output/daily-summary.part --output public/bigdata-reports/daily-summary.json
```

```
node scripts/bigdata/convert-mapreduce-output-to-json.mjs --type rain-events --input hadoop/output/rain-events.part --output public/bigdata-reports/rain-events.json
```

4.9 Lihat hasil di Web

```
npm run dev
```

Buka:

<http://localhost:3000/analytics>

<http://localhost:3000/big-data>

5. Requirement User-Friendly untuk Fitur Analysis

5.1 /analytics

- Tambahkan section atau tab: Hadoop Batch Analytics.
- Tampilkan ringkasan hasil MapReduce, bukan command Hadoop mentah.
- Sediakan tombol Open Hadoop Report ke /big-data.
- Jangan rewrite total analytics operasional yang sudah ada.

Title: Hadoop Batch Analytics

Copy: Daily MapReduce summaries generated from exported Firestore telemetry.

Cards: Total Batch Records, Device Count, Average Temperature, Rainiest Hour, Rain Events, Open vs Closed

CTA: Open Hadoop Report

5.2 /big-data

- Header: Hadoop Batch Analytics.
- Badge: Sample data atau Generated report.
- Cards: total records, device count, average temperature, rain count, rainiest hour, open vs closed.
- Daily summary table.
- Rain event aggregation table.
- Pipeline explanation.
- Cluster note: NameNode + 3 DataNodes over ZeroTier.

Pipeline copy:

Firestore sensor_data -> CSV export -> SCP over ZeroTier -> HDFS -> MapReduce -> JSON report -> Analysis UI

Runtime note:

Hadoop is used for batch processing. The Next.js app does not require Hadoop at runtime.

6. Dokumentasi yang Wajib Dibuat

File	Isi utama
docs/HADOOP_ZEROTIER_CLUSTER_RUNBOOK.md	Cluster roles, ZeroTier IP template, /etc/hosts, SSH, HDFS/YARN validation, SCP, hdfs put, run jobs, fetch output, troubleshooting.
docs/HADOOP_DATA_DICTIONARY.md	Input CSV fields, output fields, malformed row handling, date/time assumptions, Firestore mapping.
docs/HADOOP_DEMO_SCRIPT.md	Demo flow: ZeroTier, SSH, report, generate CSV, SCP, HDFS, jobs, part output, convert JSON, open UI.
docs/BIG_DATA_ROADMAP.md	Update status MVP branch dan daftar deliverables.

7. Tests dan Validasi

7.1 Web validation

npm run lint

npm run typecheck

npm run test:telegram

npm run test:architecture

npm run test:bigdata

npm run build

npm run validate

7.2 Big Data local validation

npm run bigdata:sample

npm run bigdata:local

7.3 Hadoop build

npm run bigdata:build

Jika Maven tidak ada, catat sebagai not run locally dan build di NameNode.

7.4 Cluster validation

```
ssh $HADOOP_NAMENODE_USER@$HADOOP_NAMENODE_HOST "jps && hdfs dfsadmin -report && yarn node -list"
```

7.5 Big Data contract test

- sample CSV generator exists.
- generated CSV header matches final contract.
- converter parses daily summary output.
- converter parses rain event output.
- sample JSON files are valid.
- BigDataReportService.ts exists.
- /big-data does not import fs, child_process, or Hadoop runtime.
- /analytics contains Hadoop Big Data entry point.
- Hadoop Java files exist.
- Hadoop docs exist.

Tests tidak boleh butuh live Hadoop, ZeroTier, SSH, Firebase, MQTT, Telegram, Maven, atau Java.

8. Definisi Selesai

- Web app bisa npm run validate.
- Branch feature/analysis-hadoop-big-data sudah dibuat.
- test:bigdata pass offline.
- Hadoop cluster punya Live DataNodes = 3.
- WordCount pass.
- sample CSV bisa masuk HDFS.
- DailySensorSummaryJob jalan.
- RainEventAggregationJob jalan.
- output part-r-00000 muncul.
- output bisa convert ke JSON.
- /analytics menampilkan Big Data section.
- /big-data menampilkan daily summary + rain events.
- App tetap bisa jalan tanpa Hadoop runtime.

9. Troubleshooting Cepat

Masalah	Penyebab umum	Solusi
DataNode tidak live	Hostname/hosts salah, SSH gagal, DataNode tidak start.	Cek /etc/hosts, ssh hadoop@worker hostname, start-dfs.sh, jps.
NameNode UI tidak bisa dibuka	Firewall atau hostname tidak resolve dari host.	Buka via IP ZeroTier: http://IP:9870. Allow firewall di jaringan ZeroTier.
Hadoop output already exists	Output directory sudah ada.	hdfs dfs -rm -r -f /output/smart-clothesline/<job>.
npm run bigdata:build gagal	Maven tidak ada.	Install Maven atau build di NameNode. Jangan masukkan ke validate.
App gagal build karena fs import	Client page import Node API.	Pindahkan fs/child_process ke script Node, jangan ke React page.
Converter tidak menemukan part-r-00000	Output file bernama part-00000.	Buat fetch/converter fleksibel mencari part-r-00000 dan part-00000.

10. Kalimat Penjelasan ke Dosen

Realtime control tetap memakai MQTT karena butuh respon cepat untuk device. Firestore menyimpan data historis dari sensor. Hadoop tidak digunakan untuk realtime control, tetapi untuk batch analytics. Data dari Firestore diekspor menjadi CSV, dikirim ke NameNode melalui ZeroTier, dimasukkan ke HDFS, diproses menggunakan MapReduce, lalu output part-r-00000 dikonversi menjadi JSON agar bisa ditampilkan di fitur Analysis.

Komponen	Peran
MQTT	Realtime telemetry dan command.
Firestore	Cloud storage untuk historical sensor_data.
Hadoop HDFS	Distributed storage untuk input batch analytics.
MapReduce	Distributed batch processing.
ZeroTier	Private network agar laptop/VM tim bisa mengakses cluster.
Next.js Analysis	User-friendly report view untuk hasil batch analytics.

11. Catatan Sumber Repo yang Dipakai

- package.json develop: belum ada script bigdata:*; validate masih lint/typecheck/test telegram/test architecture/build.
- src/app/analytics/page.tsx develop: sudah ada operational analytics.
- src/app/big-data/page.tsx develop: masih sample page yang fetch daily-summary.sample.json.
- docs/BIG_DATA_ROADMAP.md develop: roadmap canonical Firestore sensor_data -> CSV/HDFS/MapReduce/JSON.
- scripts/bigdata/export-firestore-sensor-data.mjs develop: masih scaffold/sample mode dan Firebase Admin export TODO.